

MEDICAL INSURANCE DATA ANALYSIS PROJECT

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Setting a working Directory

```
setwd("C:/Users/admin/Desktop/R Programming")
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats 1.0.1 v readr 2.2.0
```

```
## v ggplot2 4.0.3 v stringr 1.6.0
```

```
## v lubridate 1.9.5 v tibble 3.3.1
```

```
## v purrr 1.2.2 v tidyr 1.3.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(rio)
```

Data Importation

```
gss <- read.csv("insurance_data.csv")
head(gss)
```

```
##   age    sex    bmi children smoker   region   charges
## 1  19 female 27.900         0    yes southwest 16884.924
## 2  18  male 33.770         1    no  southeast  1725.552
## 3  28  male 33.000         3    no  southeast  4449.462
## 4  33  male 22.705         0    no northwest 21984.471
## 5  32  male 28.880         0    no northwest  3866.855
## 6  31 female 25.740         0    no  southeast  3756.622
```

```
str(gss)
```

```
## 'data.frame':   1338 obs. of  7 variables:
## $ age      : int  19 18 28 33 32 31 46 37 37 60 ...
## $ sex      : chr  "female" "male" "male" "male" ...
## $ bmi      : num  27.9 33.8 33 22.7 28.9 ...
## $ children: int  0 1 3 0 0 0 1 3 2 0 ...
## $ smoker   : chr  "yes" "no" "no" "no" ...
## $ region   : chr  "southwest" "southeast" "southeast" "northwest" ...
## $ charges  : num  16885 1726 4449 21984 3867 ...
```

Missing Values

```
miss_val <- sum(is.na(gss))
miss_val
```

```
## [1] 0
```

```
#duplicated(gss)
```

```
sum(duplicated(gss))
```

```
## [1] 1
```

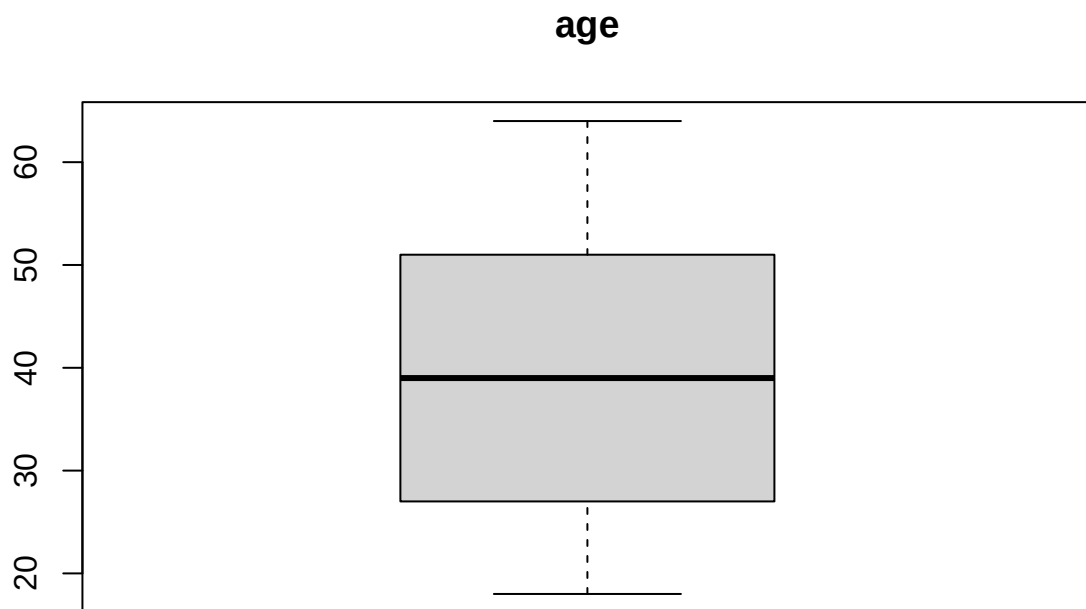
```
gss <- gss[!duplicated(gss),]
```

```
sum(duplicated(gss))
```

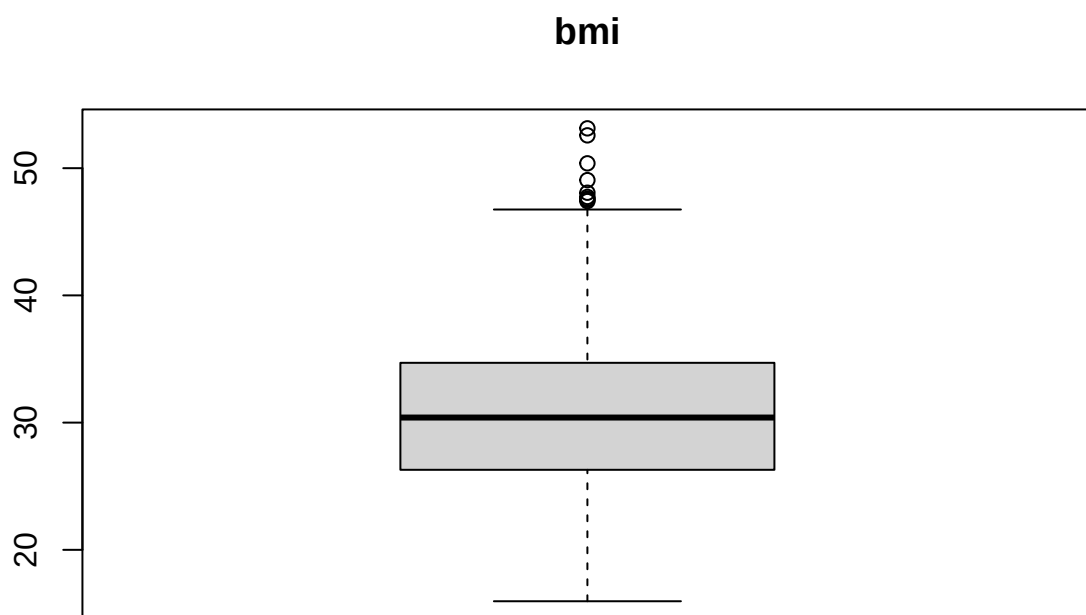
```
## [1] 0
```

Detecting Outliers

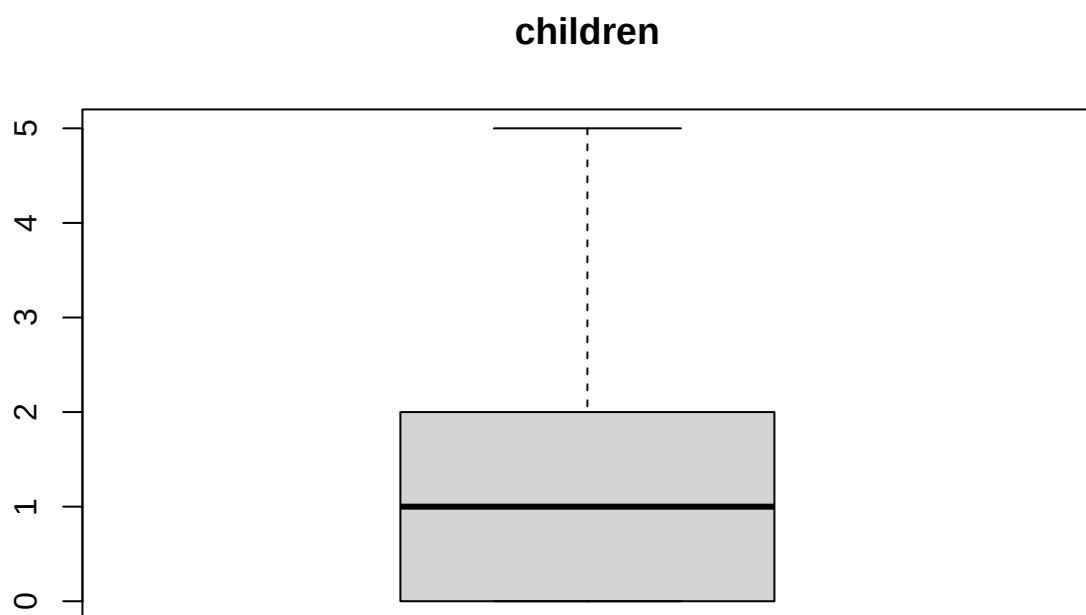
```
boxplot(gss$age, main = "age")
```



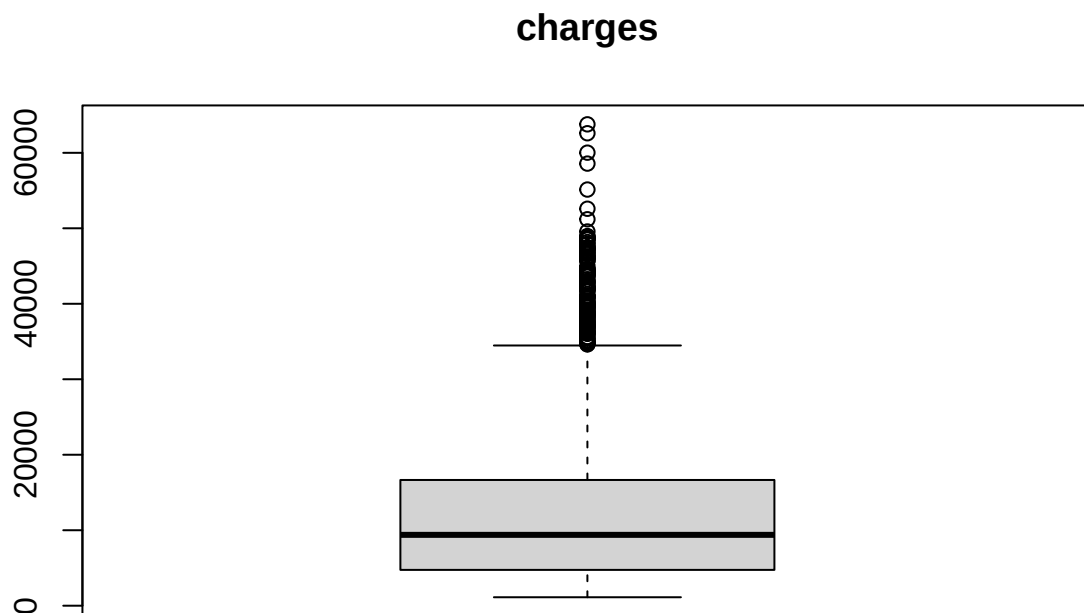
```
boxplot(gss$bmi, main = "bmi")
```



```
boxplot(gss$children, main = "children")
```



```
boxplot(gss$charges, main = "charges")
```

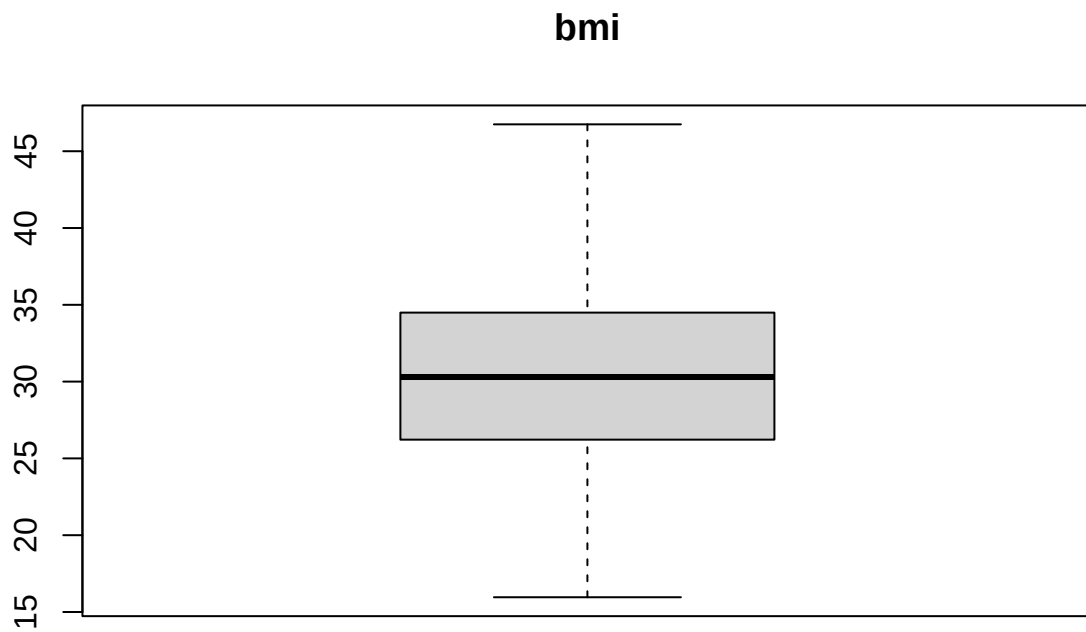


Removing Outliers

```
Q1 <- quantile(gss$bmi, 0.25)
Q3 <- quantile(gss$bmi, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

df <- gss[gss$bmi >= lower_bound & gss$bmi <= upper_bound, ]
```

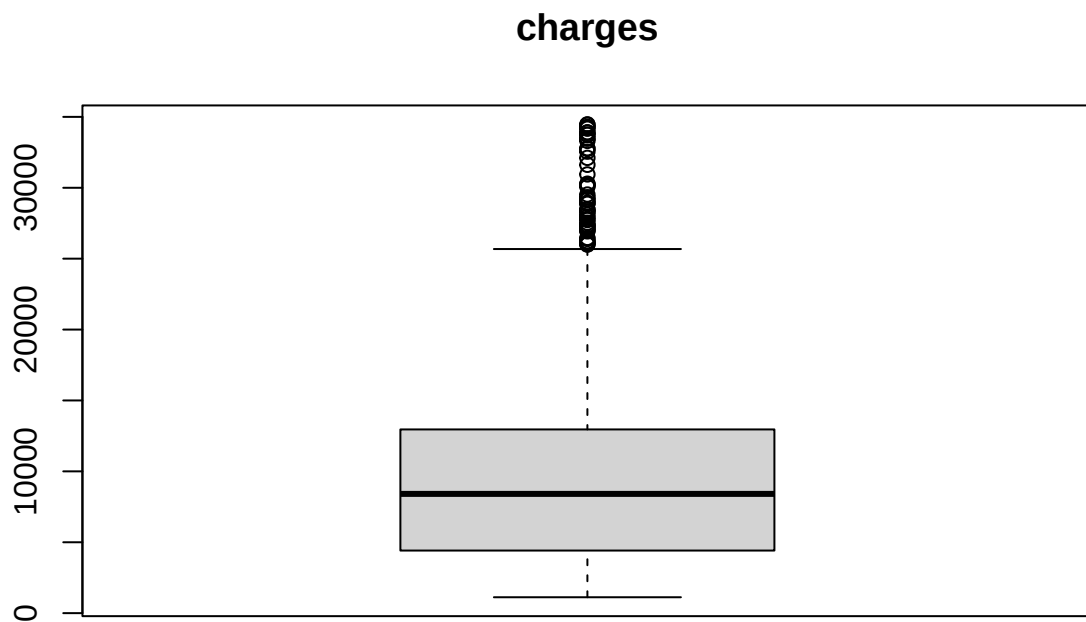
```
boxplot(df$bmi, main = "bmi")
```



```
Q1 <- quantile(gss$charges, 0.25)
Q3 <- quantile(gss$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

df <- gss[gss$charges >= lower_bound & gss$charges <= upper_bound, ]

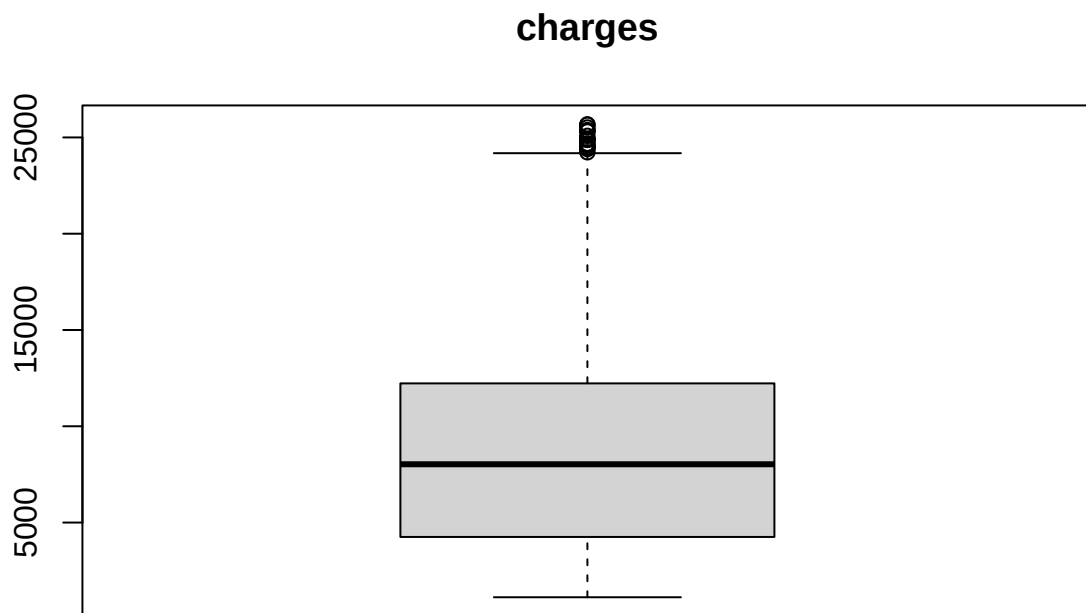
boxplot(df$charges, main = "charges")
```



```
Q1 <- quantile(df$charges, 0.25)
Q3 <- quantile(df$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

ds <- df[df$charges >= lower_bound & df$charges <= upper_bound, ]
```

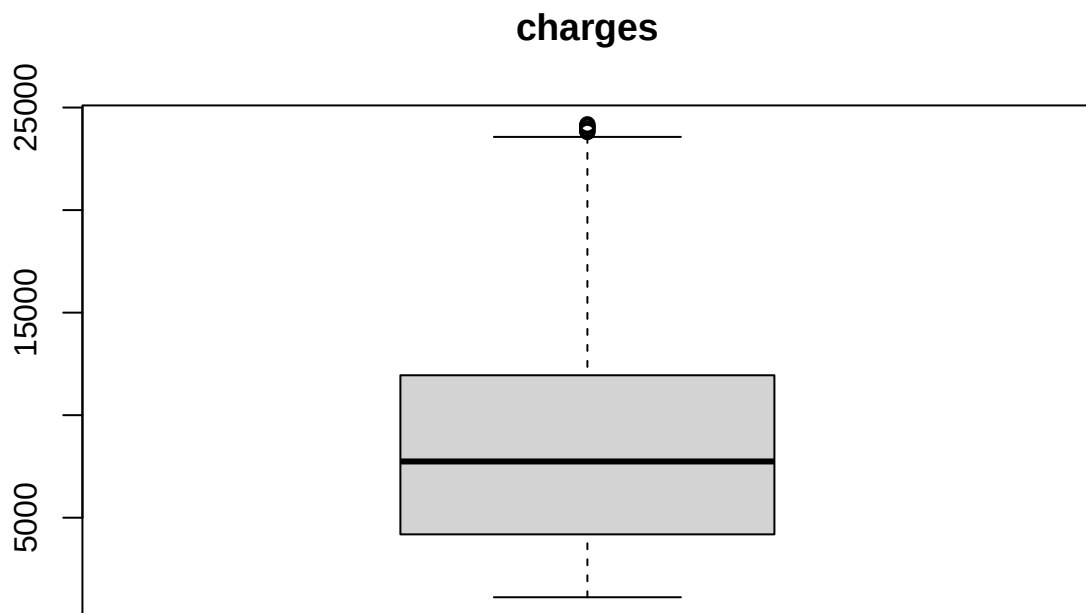
```
boxplot(ds$charges, main = "charges")
```

```
Q1 <- quantile(ds$charges, 0.25)
Q3 <- quantile(ds$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

df1 <- ds[ds$charges >= lower_bound & ds$charges <= upper_bound, ]

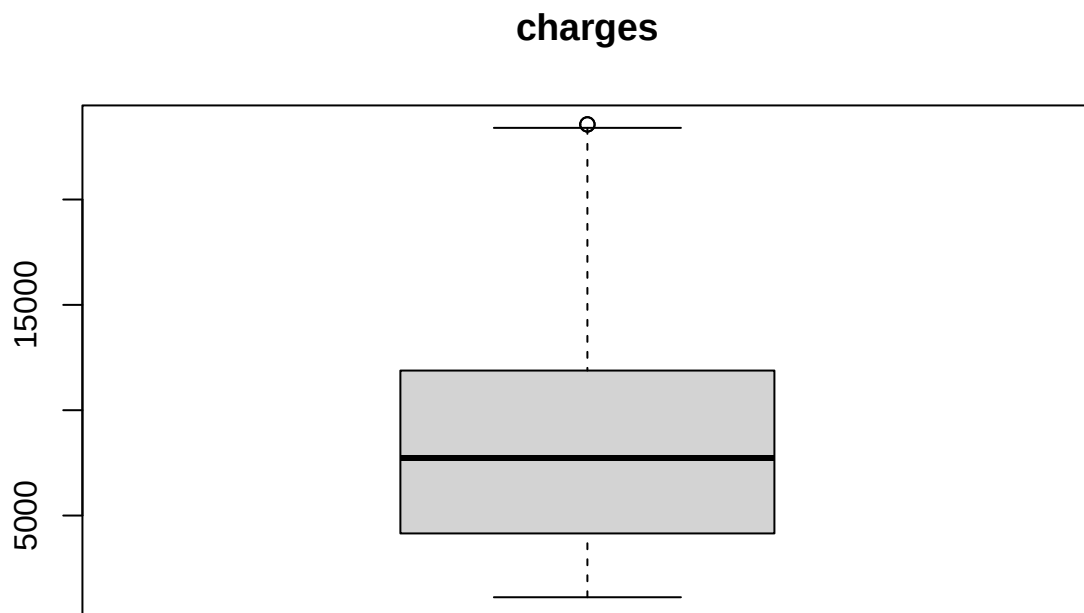
boxplot(df1$charges, main = "charges")
```



```
Q1 <- quantile(df1$charges, 0.25)
Q3 <- quantile(df1$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

df2 <- df1[df1$charges >= lower_bound & df1$charges <= upper_bound, ]

boxplot(df2$charges, main = "charges")
```

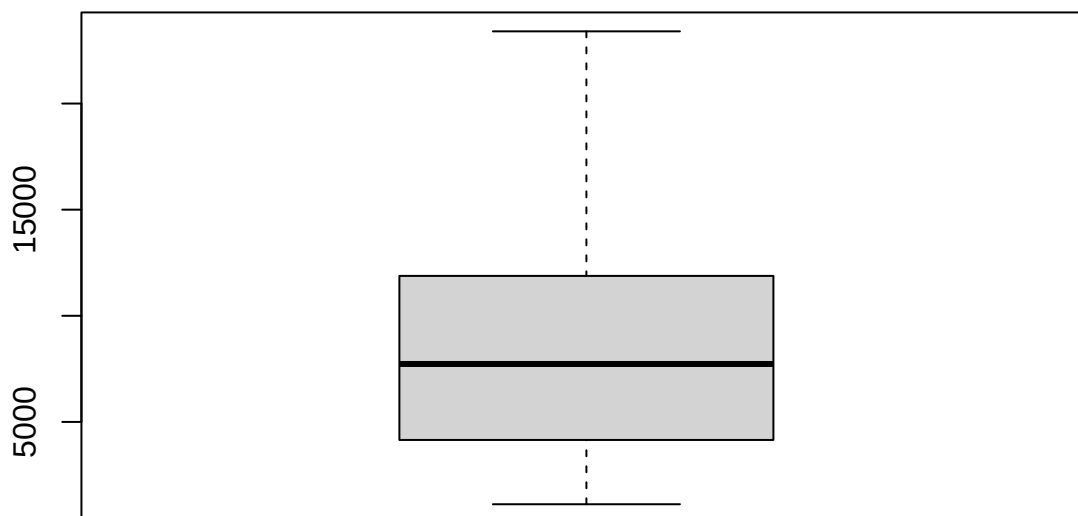


```
Q1 <- quantile(df2$charges, 0.25)
Q3 <- quantile(df2$charges, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

ds1 <- df2[df2$charges >= lower_bound & df2$charges <= upper_bound, ]

boxplot(ds1$charges, main = "charges")
```

charges



```
summary(ds1)
```

```
##      age      sex      bmi      children
##  Min.   :18.00  Length :1115  Min.    :15.96  Min.    :0.000
##  1st Qu.:26.00  N.unique :  2    1st Qu.:25.80  1st Qu.:0.000
##  Median :38.00  N.blank  :  0    Median :29.81  Median :1.000
##  Mean   :38.48  Min.nchar:  4    Mean   :30.16  Mean   :1.079
##  3rd Qu.:50.00  Max.nchar:  6    3rd Qu.:33.99  3rd Qu.:2.000
##  Max.    :64.00                      Max.    :53.13  Max.    :5.000
##      smoker      region      charges
##  Length :1115  Length :1115  Min.    : 1122
##  N.unique :  2  N.unique :  4  1st Qu.: 4150
##  N.blank  :  0  N.blank  :  0  Median : 7730
##  Min.nchar:  2  Min.nchar:  9  Mean   : 8590
##  Max.nchar:  3  Max.nchar:  9  3rd Qu.:11880
##                      Max.    :23401
```

```
summary(ds1$sex, na.rm = TRUE)
```

```
##      Length  N.unique  N.blank  Min.nchar  Max.nchar
##      1115         2         0         4         6
```

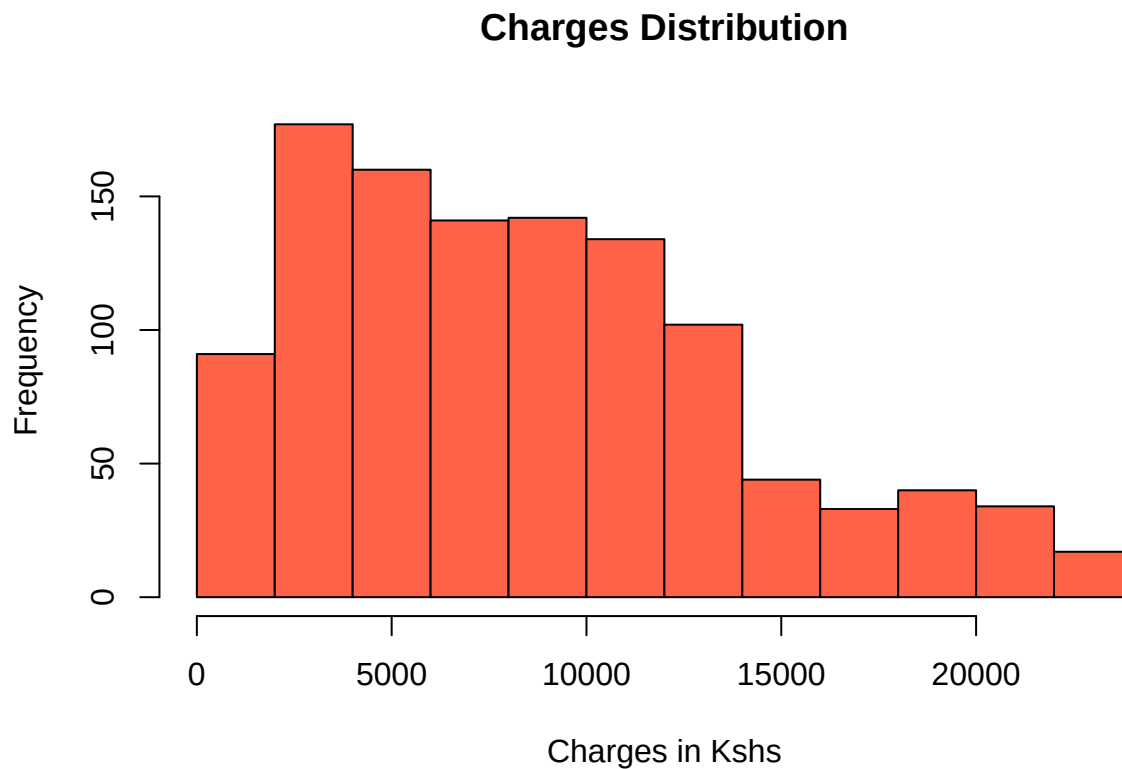
```
mode_value<-names(sort(table(ds1$region), decreasing = TRUE))[1]
mode_value
```

```
## [1] "southeast"
```

EDA(Exploratory Data Analysis)

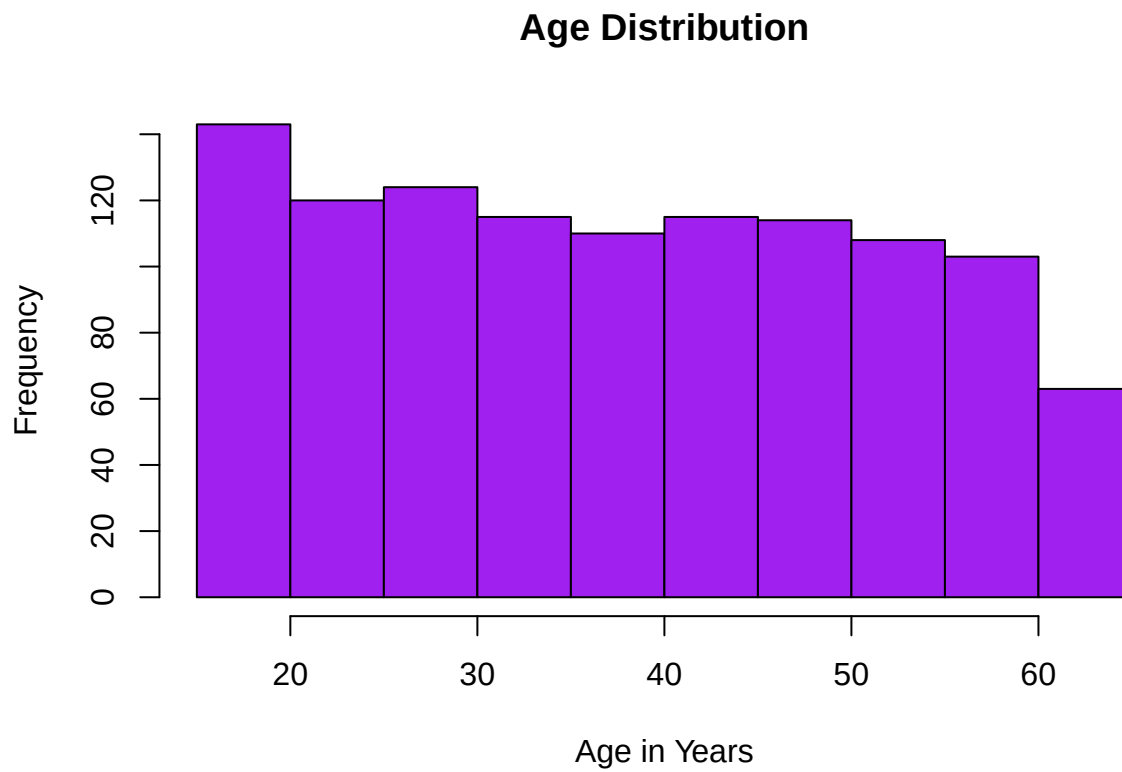
Histogram

```
hist(ds1$charges,  
     main = "Charges Distribution",  
     xlab = "Charges in Kshs",  
     col = "tomato")
```



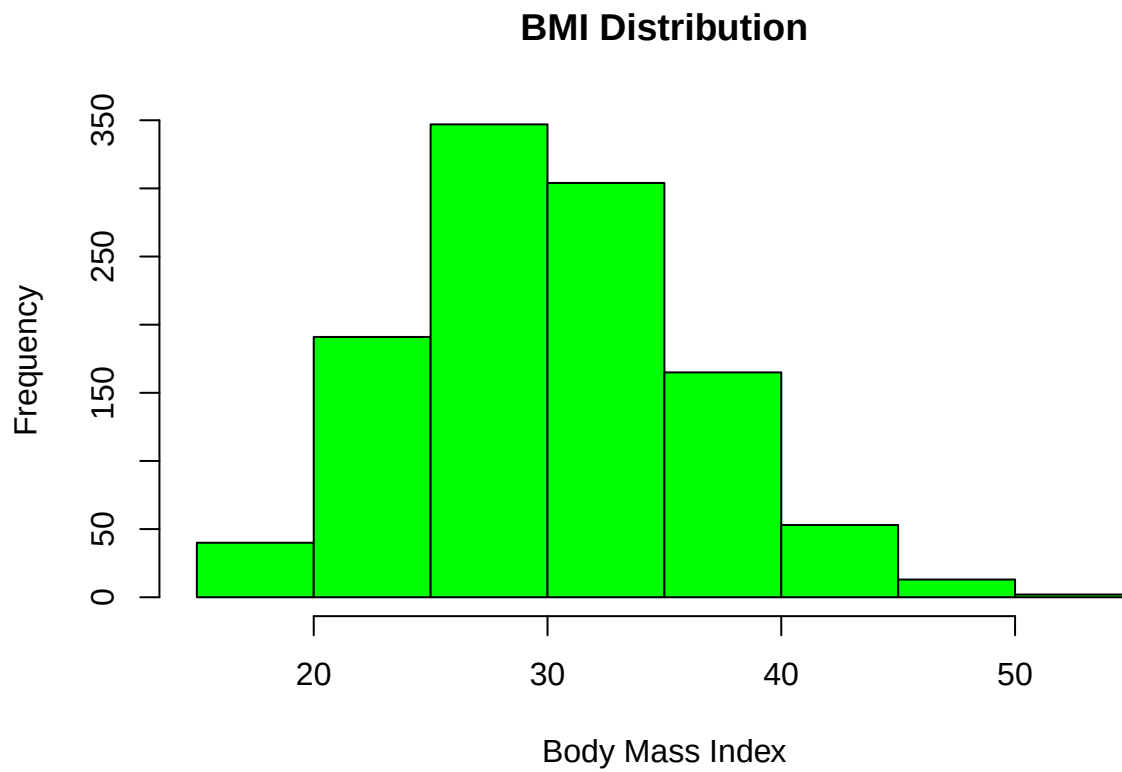
> Charges Distribution has a Positively Skewed distribution.

```
hist(ds1$age,  
     main = "Age Distribution",  
     xlab = "Age in Years",  
     col = "purple")
```



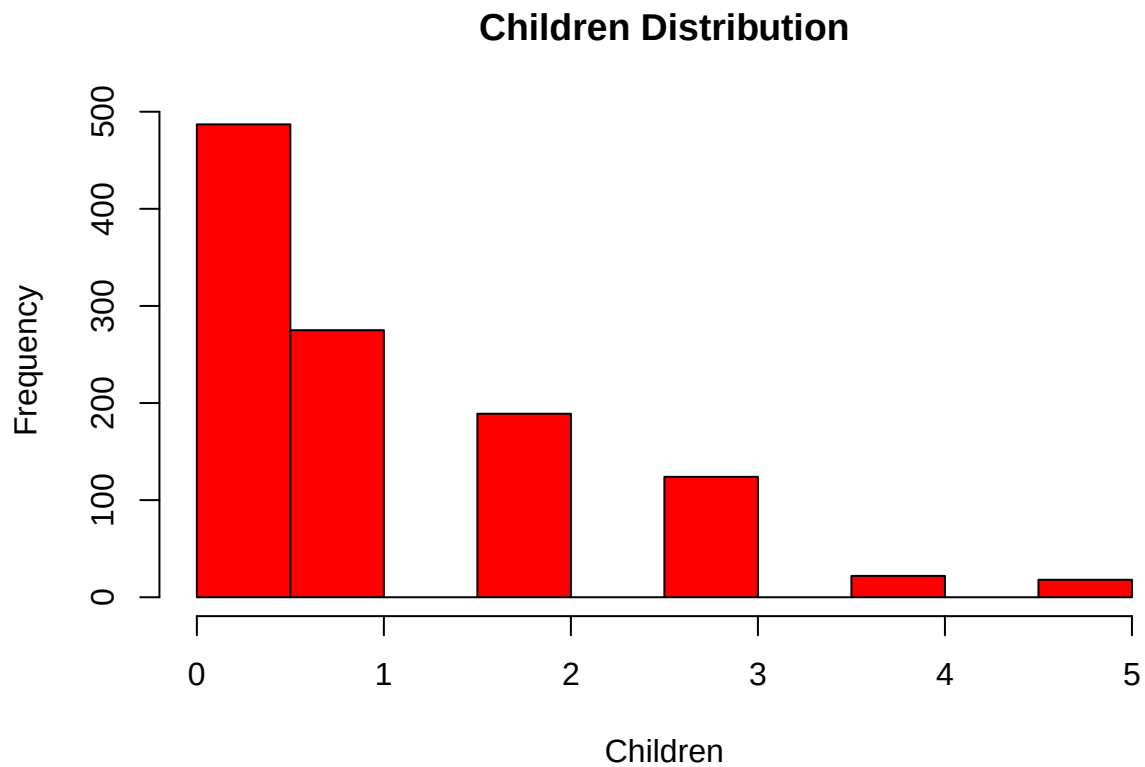
Age Distribution has a uniform distribution since the frequencies across age groups are fairly similar.

```
hist(ds1$bmi,  
     main = "BMI Distribution",  
     xlab = "Body Mass Index",  
     col = "green")
```



Body Mass Index Distribution has a normal distribution.

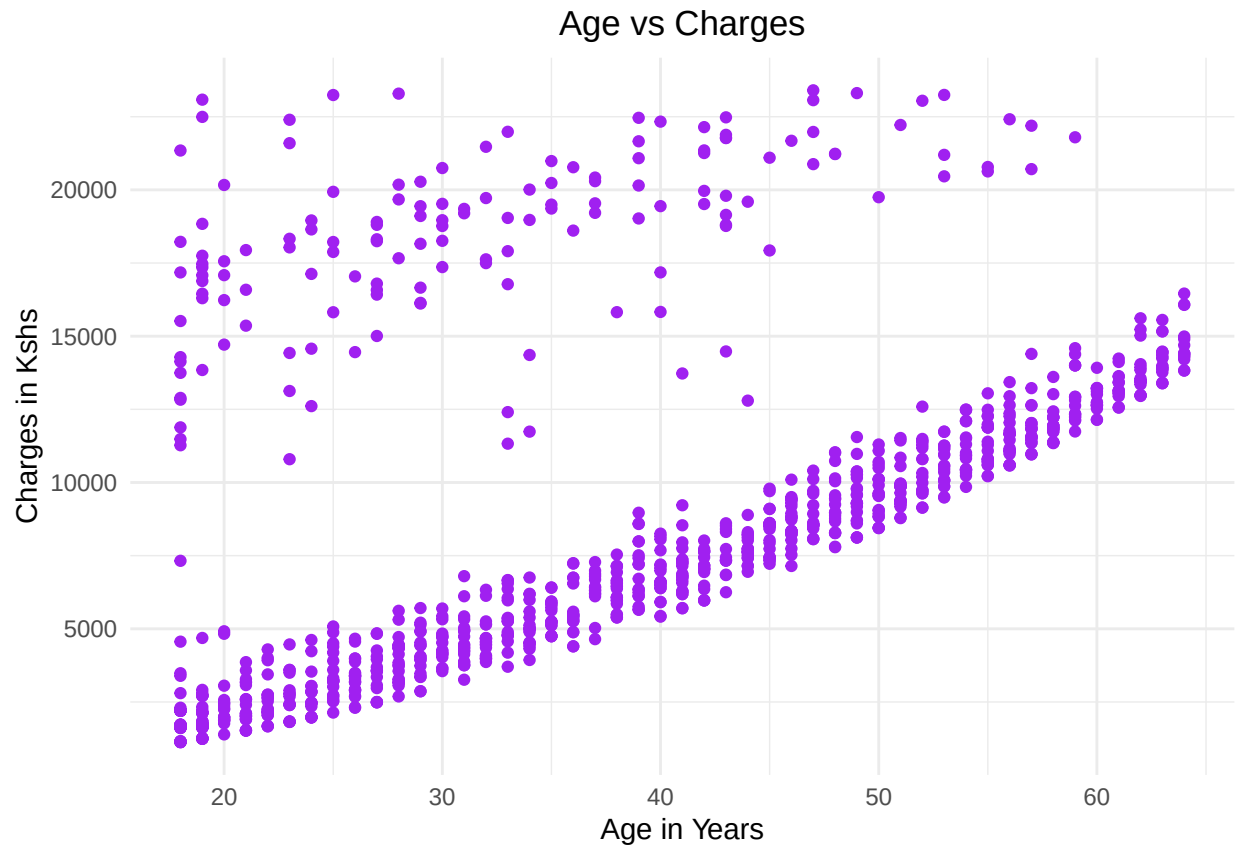
```
hist(ds1$children,  
     main = "Children Distribution",  
     xlab = "Children",  
     col = "red")
```



Children Distribution has a positively skewed distribution since majority of individuals has small number of children.

Scatter Plot

```
library(ggplot2)
ggplot(ds1, aes(x = age, y = charges)) +
  geom_point(color = "purple") +
  labs(title = "Age vs Charges", x = "Age in Years", y = "Charges in Kshs") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

The scatter plot indicates that age is a strong predictor of insurance charges implying as age increases charges also increases. It shows a clear positive relationship between age and charges.

Data Manipulation

Filter

```
ds1_filter <- ds1 |>
  dplyr::filter(age>30)
head(ds1_filter)
```

```
##   age    sex    bmi children smoker   region  charges
## 1  33  male  22.705         0    no northwest 21984.471
## 2  32  male  28.880         0    no northwest  3866.855
## 3  31 female  25.740         0    no southeast  3756.622
## 4  46 female  33.440         1    no southeast  8240.590
## 5  37 female  27.740         3    no northwest  7281.506
## 6  37  male  29.830         2    no northeast  6406.411
```

```
ds1_c <- ds1 |>
  dplyr::filter(smoker=='yes')
head(ds1_c)
```

```
##   age    sex    bmi children smoker    region  charges
## 1  19 female 27.900         0    yes southwest 16884.92
## 2  53 female 22.880         1    yes southeast 23244.79
## 3  20 female 22.420         0    yes northwest 14711.74
## 4  28  male 23.980         3    yes southeast 17663.14
## 5  27 female 24.750         0    yes southeast 16577.78
## 6  45  male 22.895         2    yes northwest 21098.55
```

```
ds1_t <- ds1 |>
  dplyr::filter(region=='northwest')
head(ds1_t)
```

```
##   age    sex    bmi children smoker    region  charges
## 1  33  male 22.705         0    no northwest 21984.471
## 2  32  male 28.880         0    no northwest  3866.855
## 3  37 female 27.740         3    no northwest  7281.506
## 4  37  male 28.025         2    no northwest  6203.902
## 5  55 female 32.775         2    no northwest 12268.632
## 6  23  male 17.385         1    no northwest  2775.192
```

Select

```
ds1_select <- ds1 |>
  dplyr::select(sex, charges, age)
head(ds1_select)
```

```
##      sex    charges age
## 1 female 16884.924 19
## 2  male  1725.552 18
## 3  male  4449.462 28
## 4  male 21984.471 33
## 5  male  3866.855 32
## 6 female  3756.622 31
```

Mutate

```
ds1_mutate <- ds1|>
  mutate(charges_category = ifelse(charges > 10000, 'High', 'Low'))
head(ds1_mutate)
```

```
##   age    sex    bmi children smoker    region  charges charges_category
## 1  19 female 27.900         0    yes southwest 16884.924          High
## 2  18  male 33.770         1    no southeast  1725.552           Low
## 3  28  male 33.000         3    no southeast  4449.462           Low
## 4  33  male 22.705         0    no northwest 21984.471          High
## 5  32  male 28.880         0    no northwest  3866.855           Low
## 6  31 female 25.740         0    no southeast  3756.622           Low
```

Arrange

```
ds1_arrange <- ds1|>
  arrange(desc(charges))
head(ds1_arrange, 20)
```

```
##   age  sex  bmi children smoker  region  charges
## 1  47 female 26.125      1   yes northeast 23401.31
## 2  49  male 25.600      2   yes southwest 23306.55
## 3  28 female 24.320      1   no  northeast 23288.93
## 4  53 female 22.880      1   yes southeast 23244.79
## 5  25  male 24.985      2   no  northeast 23241.47
## 6  19  male 33.100      0   no  southwest 23082.96
## 7  47 female 27.830      0   yes southeast 23065.42
## 8  52 female 30.875      0   no  northeast 23045.57
## 9  19  male 27.265      2   no  northwest 22493.66
## 10 43 female 26.700      2   yes southwest 22478.60
## 11 39  male 29.925      1   yes northeast 22462.04
## 12 56  male 19.950      0   yes northeast 22412.65
## 13 23 female 24.225      2   no  northeast 22395.74
## 14 40 female 28.120      1   yes northeast 22331.57
## 15 51  male 23.210      1   yes southeast 22218.11
## 16 57 female 23.980      1   no  southeast 22192.44
## 17 42  male 30.000      0   yes southwest 22144.03
## 18 33  male 22.705      0   no  northwest 21984.47
## 19 47  male 25.410      1   yes southeast 21978.68
## 20 43 female 24.700      2   yes northwest 21880.82
```

Grouping

```
ds1|>
  group_by(sex) |>
  summarize(mean_charges = mean(charges, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   sex      mean_charges
##   <chr>          <dbl>
## 1 female          8762.
## 2 male            8411.
```

```
ds1|>
  group_by(smoker) |>
  summarize(mean_charges = mean(charges, na.rm = TRUE))
```

```
## # A tibble: 2 x 2
##   smoker mean_charges
##   <chr>          <dbl>
## 1 no            7660.
## 2 yes          18814.
```

```
ds1|>
  group_by(region) |>
  summarize(mean_bmi = mean(bmi, na.rm = TRUE)) |>
  arrange(desc(mean_bmi))
```

```
## # A tibble: 4 x 2
##   region    mean_bmi
##   <chr>      <dbl>
## 1 southeast    32.9
## 2 southwest    30.2
## 3 northwest    28.8
## 4 northeast    28.7
```

```
ds1_processed <- ds1|>
  filter(age > 30) |>
  select(region, charges, age) |>
  mutate(age_category = ifelse(age > 28, "Adults", "Minor")) |>
  arrange(desc(charges))
head(ds1_processed, 20)
```

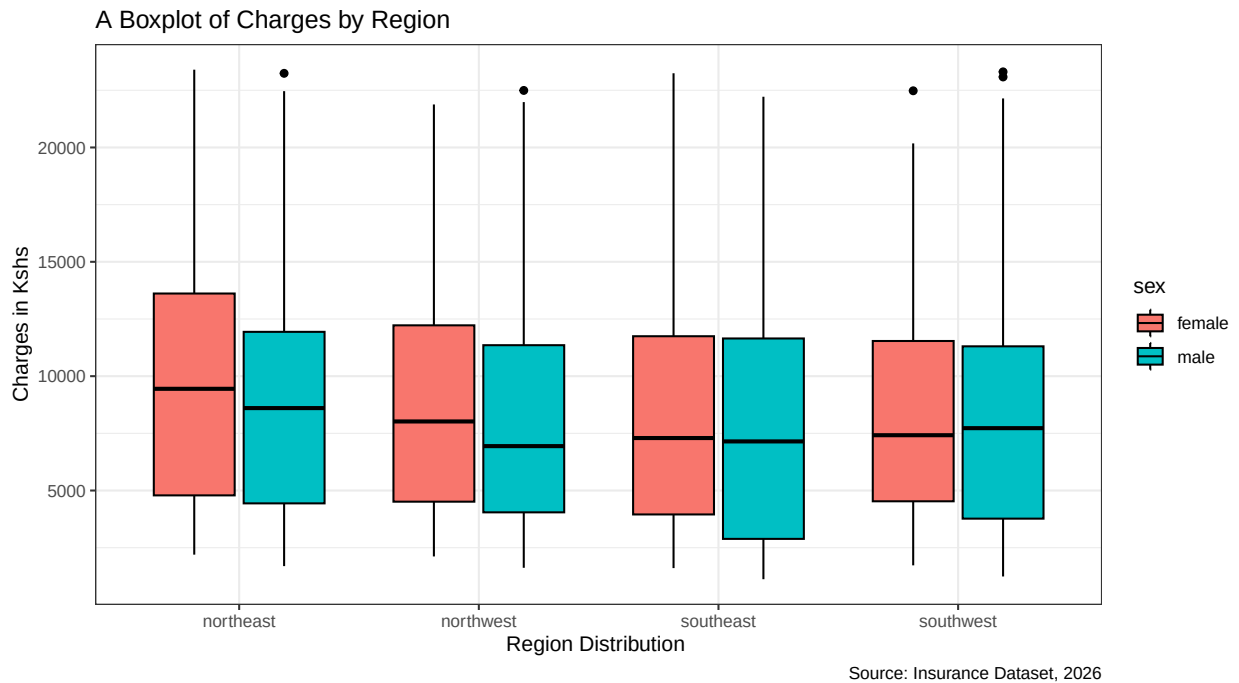
```
##      region  charges age age_category
## 1  northeast 23401.31  47      Adults
## 2  southwest 23306.55  49      Adults
## 3  southeast 23244.79  53      Adults
## 4  southeast 23065.42  47      Adults
## 5  northeast 23045.57  52      Adults
## 6  southwest 22478.60  43      Adults
## 7  northeast 22462.04  39      Adults
## 8  northeast 22412.65  56      Adults
## 9  northeast 22331.57  40      Adults
## 10 southeast 22218.11  51      Adults
## 11 southeast 22192.44  57      Adults
## 12 southwest 22144.03  42      Adults
## 13 northwest 21984.47  33      Adults
## 14 southeast 21978.68  47      Adults
## 15 northwest 21880.82  43      Adults
## 16 southwest 21797.00  59      Adults
## 17 northwest 21774.32  43      Adults
## 18 northeast 21771.34  43      Adults
## 19 northwest 21677.28  46      Adults
## 20 northeast 21659.93  39      Adults
```

Data Visualization

Box Plot

```
library(ggplot2)
ds1 |>
  ggplot(aes(x = region, y = charges, fill = sex))+
  geom_boxplot(color = "black")+
```

```
labs(x = "Region Distribution ",
     y = "Charges in Kshs",
     title = "A Boxplot of Charges by Region",
     caption = "Source: Insurance Dataset, 2026")+
theme_bw()
```



The graph shows that insurance varies across all regions between gender. Southeast region has the highest median charges for both sexes, while the northwest and northeast regions have relatively lower and similar distributions. Male charges tend to be slightly higher than female charges across the regions. There are also few outliers indicating some individuals have high insurance charges.

```
library(plotly)
```

```
##
## Attaching package: 'plotly'

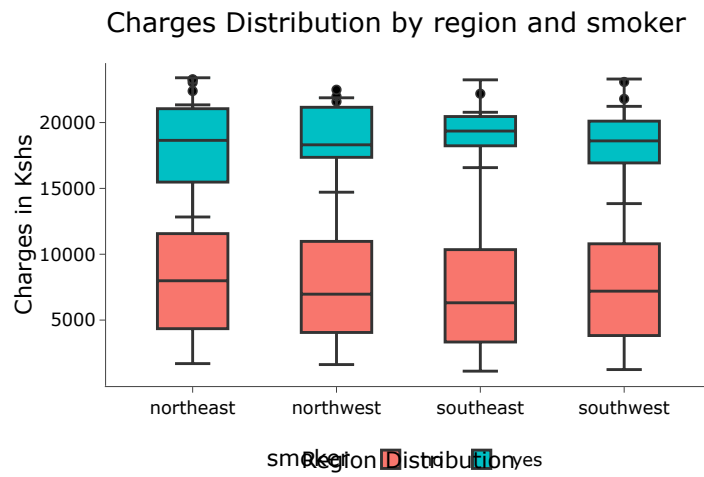
## The following object is masked from 'package:rio':
##
##   export

## The following object is masked from 'package:ggplot2':
##
##   last_plot

## The following object is masked from 'package:stats':
##
##   filter
```

```
## The following object is masked from 'package:graphics':  
##  
## layout
```

```
p = ds1|>  
  ggplot(aes(x = region, y= charges, fill = smoker)) +  
  geom_boxplot() +  
  labs(title = "Charges Distribution by region and smoker",  
        x = "Region Distribution",  
        y = "Charges in Kshs",  
        caption = "Source: Insurance Dataset, 2026")+  
  theme_classic()+  
  theme(legend.position = "bottom")  
ggplotly(p)
```



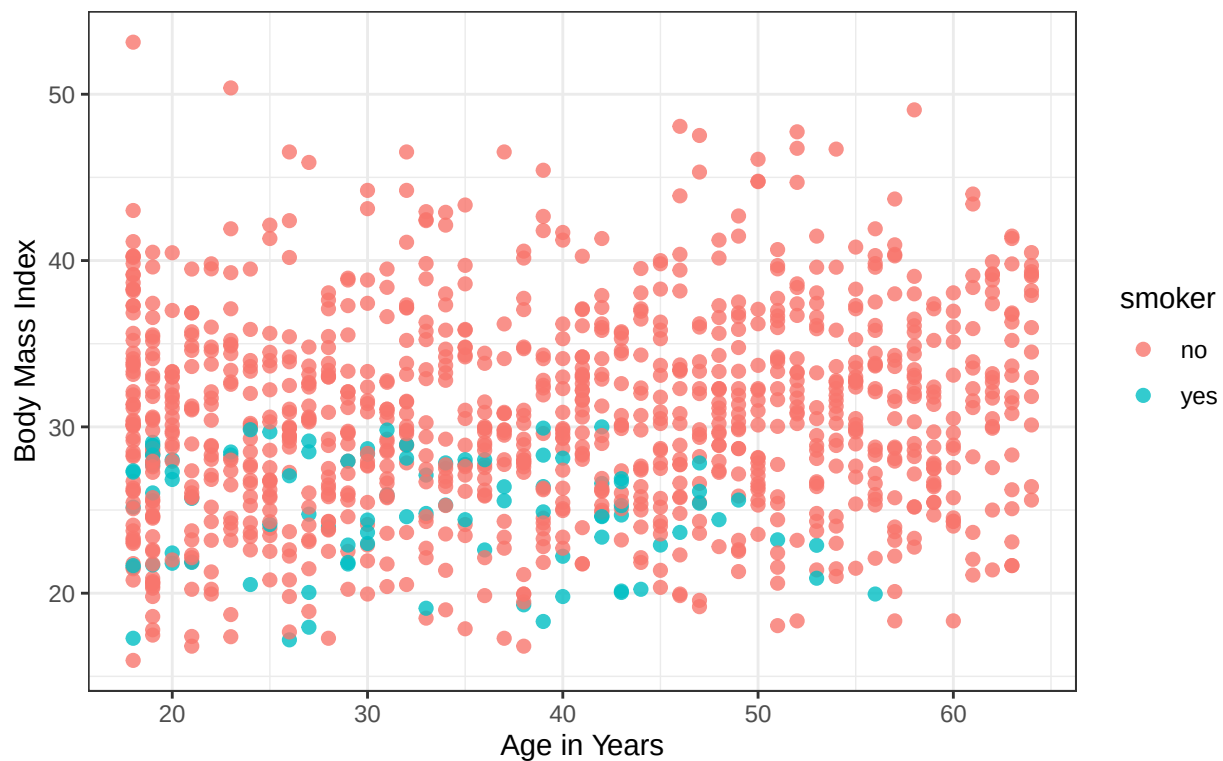
This indicates that insurance charges vary across all regions but a quite similar distribution. Smokers tend to have higher charges than nono smokers in every region as seen using the median

values. There also exist outliers which show that some individuals have extremely high insurance charges.

Scatter Plot

```
ds1 |> ggplot(aes(x = age, y = bmi, color = smoker))+  
  geom_point(size = 2, alpha = 0.8)+  
  labs(x = "Age in Years",  
       y = "Body Mass Index",  
       title = "Scatter plot showing relationship between age and BMI",  
       caption = "Source: Insurance Dataset, 2026")+  
  theme_bw()
```

Scatter plot showing relationship between age and BMI



Source: Insurance Dataset, 2026

> Bmi values are spread across all age groups indicating there's no strong relationship between age and bmi. Most individuals have bmi values between 20 and 40 and non smokers appear to be the majority.

Bar Chart

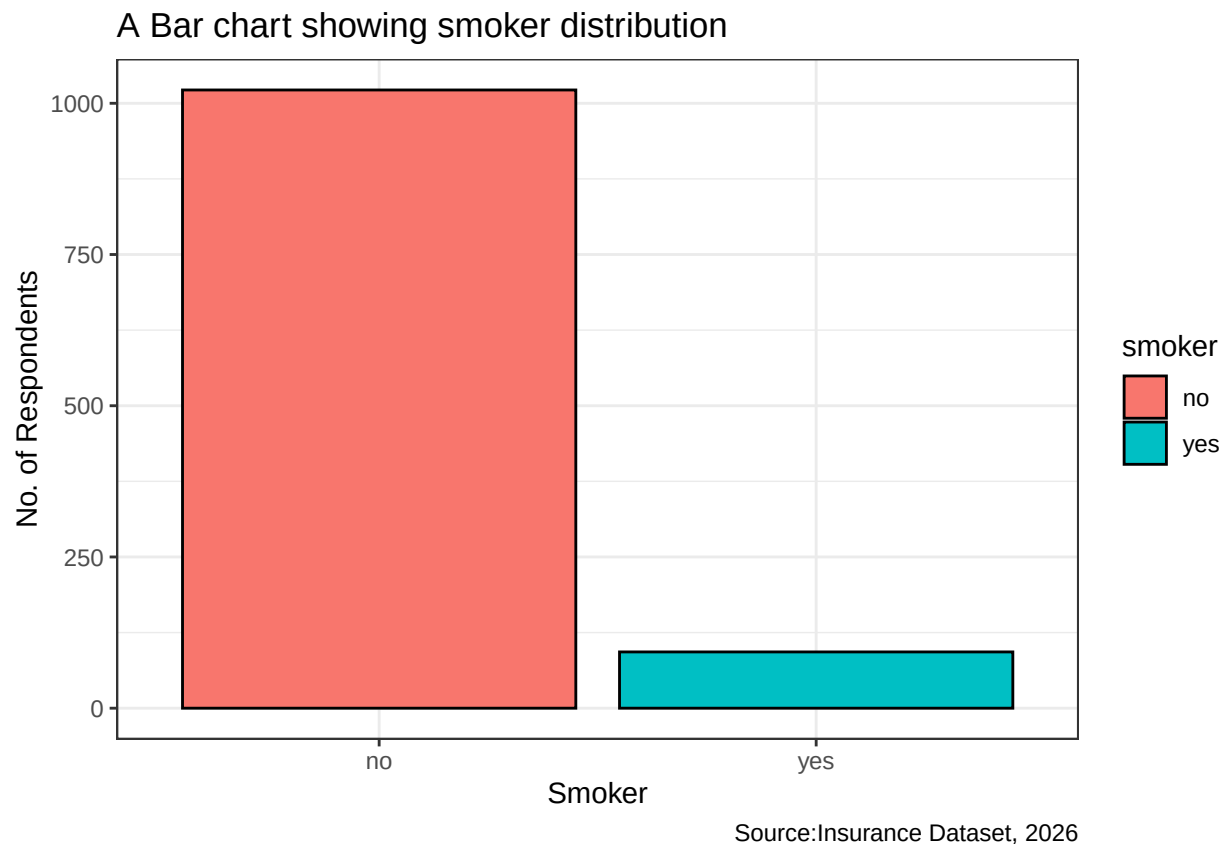
```
ds1 |>  
  ggplot(aes(x = smoker, fill = smoker))+  
  geom_bar(color = "black")+  
  labs(x = "Smoker",  
       y = "No. of Respondents",
```



```

title = "A Bar chart showing smoker distribution ",
caption = "Source:Insurance Dataset, 2026")+
theme_bw()

```



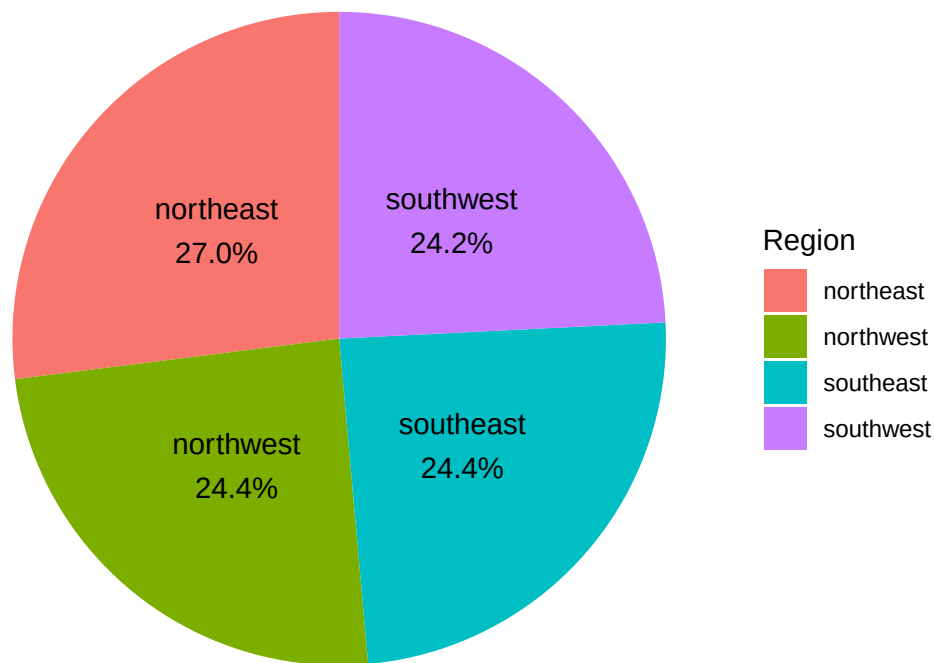
Non smokers make the largest number in the dataset.

```

ds1|>
  group_by(region)|>
  summarise(total_charges = sum(charges, na.rm = TRUE))|>
  mutate(
    pct = total_charges / sum(total_charges),
    label = paste0(region, "\n", scales::percent(pct, accuracy = 0.1))
  )|>
  ggplot(aes(x = "", y = total_charges, fill = region))+
  geom_col(width = 1)+
  geom_text(aes(label = label), position = position_stack(vjust = 0.5), size = 4)+
  coord_polar(theta = "y")+
  labs(
    title = "Total charges by region",
    fill = "Region"
  )+
  theme_void()

```

Total charges by region



The northeast Region has the highest share of total charges at 27.0%. The northwest and southeast regions contribute equally at 24.4% each. Southwest has the lowest share at 24.2%.

Hypothesis Testing

Two Sample T-Test

Ho: Average insurance charges for males is equal to average insurance charges for females. H1: Average insurance charges for males differs significantly from the average insurance charges for females.

```
result <- t.test(charges ~ sex, data = ds1)
result
```

```
##
## Welch Two Sample t-test
##
## data: charges by sex
## t = 1.0779, df = 1102.5, p-value = 0.2813
## alternative hypothesis: true difference in means between group female and group male is not equal to
## 95 percent confidence interval:
## -287.4416 988.3175
## sample estimates:
## mean in group female mean in group male
## 8761.859 8411.421
```

Characteristic	no N = 1,022 ¹	95% CI	yes N = 93 ¹	95% CI	p-value ²
sex					0.7
female	524 (51%)	48%, 54%	45 (48%)	38%, 59%	
male	498 (49%)	46%, 52%	48 (52%)	41%, 62%	

¹n (%)

²Pearson's Chi-squared test

Abbreviation: CI = Confidence Interval

From the results above, fail to reject Ho since p-value is greater than 0.05 and conclude that the average insurance charges between males and females is equal at 5% level of significance.

Ho: There is no significant difference in average body mass index between smokers and non smokers. H1: There is a significant difference in average body mass index between smokers and non smokers.

```
result <- t.test(bmi ~ smoker, data = ds1)
result

##
## Welch Two Sample t-test
##
## data:  bmi by smoker
## t = 14.438, df = 155.82, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group no and group yes is not equal to 0
## 95 percent confidence interval:
##  4.890029 6.440092
## sample estimates:
## mean in group no mean in group yes
##      30.63167      24.96661
```

Since p-value < 2.2e-16 which is less than 0.05, reject Ho at 5% level of significance and conclude that there is a significant difference in average body mass index between smokers and non smokers.

Chi-Square Test

Ho: There is no significant association between smoking status and gender. H1: There is a significant association between smoking status and gender.

```
library(gtsummary)

tbl_summary(ds1[c("sex", "smoker")],
            by = smoker) %>%
  add_ci() %>%
  add_p(
    test = list(
      all_categorical() ~ "chisq.test"
    )
  )
```

The table above indicates that p-value is greater than 0.05 therefore fail to reject H_0 at 5% level of significance and conclude that there is no significant association between smoking status and gender.

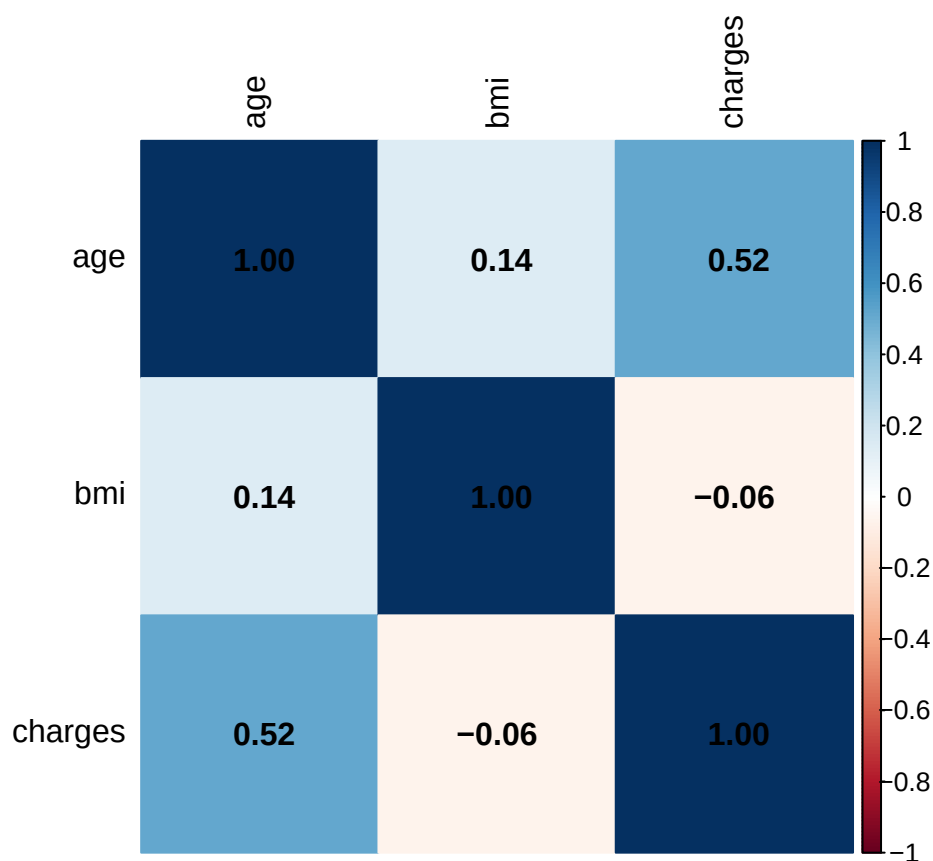
Correlation Analysis(Using Heatmap)

Heatmap

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
ds1|>  
  select(age, bmi, charges)|>  
  cor(use = "complete.obs")|>  
  corrplot(method = "color",  
           addCoef.col = "black", tl.col = "black")
```



The heatmap indicates that age has a positive relationship in both bmi and charges where in charges it has a strong positive relationship implying insurance costs rises as age increases. In bmi it has a weak positive relationship. Bmi and charges has a negative relationship suggesting that bmi alone does not strongly correlate with charges.

Regression Analysis

Simple Linear Regression Ho: Age has no significant effect on Charges. H1: Age has a significant effect on Charges.

```
model <- lm(charges~age, data = ds1)
summary(model)
```

```
##
## Call:
## lm(formula = charges ~ age, data = ds1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3809.0 -2630.4 -1654.2  -194.1 18452.7
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   768.49     409.48   1.877  0.0608 .
## age           203.25      10.01  20.302 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4633 on 1113 degrees of freedom
## Multiple R-squared:  0.2703, Adjusted R-squared:  0.2696
## F-statistic: 412.2 on 1 and 1113 DF,  p-value: < 2.2e-16
```

From the analysis above it shows that age expresses charges by 27.03% while the remaining percent is expressed by other variables which are not in this model.

As one unit of age increases it increases charges by 203.25 units. Since p-value < 2e-16 which is less than 0.05, reject Ho at 5% level of significance and conclude that age has a significant effect on charges.

Multiple Linear Regression Ho: Age, BMI, Children, smoker and gender do not have significant effect on charges. H1: At least one of the variables has a significant effect on charges.

```
model <- lm(charges ~ age + bmi + children + sex + smoker, data = ds1)
summary(model)
```

```
##
## Call:
## lm(formula = charges ~ age + bmi + children + sex + smoker, data = ds1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3622.5 -1199.9  -774.9   -35.0 20704.3
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2274.967     528.660  -4.303 1.83e-05 ***
## age          234.815       6.592   35.620 < 2e-16 ***
```

```
## bmi          15.965      15.358    1.040    0.2988
## children     409.989      73.903    5.548 3.62e-08 ***
## sexmale      -336.307     179.949   -1.869    0.0619 .
## smokeryes    12823.326    338.121   37.925 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3001 on 1109 degrees of freedom
## Multiple R-squared:  0.6949, Adjusted R-squared:  0.6935
## F-statistic: 505.1 on 5 and 1109 DF,  p-value: < 2.2e-16
```

The model expresses charges by 69.49 percentage.

As one unit of age increases it increases charges by 234.815 units. Since p-value < 2e-16 which is less than 0.05, it implies that age has a significant effect on charges.

As Body mass index increases by a unit it increases charges by 15.965 units. Since p-value is greater than 0.05, it implies that body mass index do not have a significant effect on charges.

For each additional child, charges increase by 409.989 units. Since p-value is less than 0.05, it indicates that children has a significant effect on charges.

Males have charges 336.31 units lower than females. Since p-value is greater than 0.05, it shows that gender do not have a significant effect on charges.

Smokers pay charges by 12,823.33 units more than non-smokers. Since p-value < 2e-16 which is less than 0.05, it implies that smokers have a significant effect on charges.

Thus, reject H_0 and conclude that at least one of the variables has a significant effect on charges.

**SPECIAL THANKS TO: CENTER FOR DATA ANALYTICS
AND MODELLING - CHUKA UNIVERSITY**

**CERTIFICATION COURSE: BASIC INTRODUCTION TO R
FOR DATA SCIENCE.**